Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

R26-IT-018

Project Proposal Report

Nawarathna I K K R B - IT19131948

B.Sc. (Hons) Degree in Information Technology Specialized in Computer Systems & Network Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Nawarathna D.K.K.R.B	IT 19131948	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Prof Anuradha Jayakody	13/03/2026	

ABSTRACT

Real-time web-based educational systems require secure and efficient communication to support reliable interaction between users and backend services. In the proposed emotion-aware e-learning platform for Deaf G.C.E. O/L Information Technology students, learner-related data such as interaction records, support requests, and emotion-associated information must be transmitted safely and with minimal delay. However, many web-based systems emphasize application functionality while giving limited attention to bandwidth efficiency and in-transit data protection, particularly in low-connectivity environments. This creates a research gap in secure and optimized communication frameworks for real-time educational applications.

This individual study proposes a secure and bandwidth-optimized communication framework for the broader emotion-aware e-learning web application. The solution focuses on protected client-server communication using HTTPS/TLS, payload minimization, HTTP compression techniques such as GZIP or Brotli, and basic access protection mechanisms. The framework will be evaluated experimentally by comparing baseline communication with the proposed optimized secure communication approach under normal and constrained network conditions.

The expected outcomes include reduced bandwidth consumption, improved transmission efficiency, secure handling of learner-related data, and measurable performance improvement using metrics such as latency, throughput, bandwidth usage, and security overhead. This contribution strengthens the technical reliability and deployment readiness of the broader educational platform while improving its suitability for bandwidth-constrained environments such as Sri Lanka.

Keywords: *Secure Web Communication, Bandwidth Optimization, HTTPS/TLS, Network Performance Evaluation, Educational Web Systems*

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Prof. Anuradha Jayakody, for his valuable guidance, continuous encouragement, and insightful advice throughout the preparation of this proposal. His academic direction and constructive feedback greatly helped me to shape this individual research contribution in a clear and meaningful manner.

I am also deeply thankful to my co-supervisor, Senior Lecturer Dinith Primal, for his support, suggestions, and technical guidance during the planning of this study. His observations and recommendations were highly valuable in improving the quality and focus of this proposal.

I would also like to express my appreciation to my fellow group members for their cooperation, ideas, and support throughout the early stages of this project. Their discussions and collaborative efforts helped establish the broader project context within which this individual contribution was developed.

In addition, I am grateful to the academic staff of the Department of Information Technology, Sri Lanka Institute of Information Technology, for providing the knowledge, resources, and academic foundation necessary for this work. Finally, I would like to thank my family and friends for their encouragement, patience, and continuous support throughout this process.

TABLE OF CONTENT

Contents

DECLARATION....ii
ABSTRACT.... iii
ACKNOWLEDGEMENT....iv
TABLE OF CONTENT....v
LIST OF ABBREVIATIONS....vii
1. INTRODUCTION....1
1.1 Background and Context....1
1.2 Literature Review....2
1.3 Research Gap....3
1.4 Research Problem....3
2. OBJECTIVE....4
2.1 Main Objective....4
2.2 Specific Objectives....4
3. METHODOLOGY....5
4. HIGH-LEVEL SYSTEM ARCHITECTURE....7
4.1 Overview of Proposed Solution....7
4.2 High-Level System Architecture Diagram....8
4.3 Individual Components and Functions....9
1. Secure Transport Module....9
2. Payload Optimization Module....9
3. Access Control and Protection Module....10
4. Monitoring and Evaluation Module....10
4.4 Data Flow Explanation....10
4.5 Scalability and Security Considerations....11
5. USER REQUIREMENTS....12
5.1 Stakeholder Identification....12
5.2 Functional Requirements....12
5.3 Non-Functional Requirements....13
5.3.1 Security Requirements....13
5.3.2 Performance Requirements....13
5.3.3 Reliability Requirements....13

5.3.4 Scalability Requirements 13
5.3.5 Maintainability Requirements 13
6. COMMERCIALIZATION PLAN 14
6.1 Target Market and Customer Persona 14
6.2 Value Proposition 14
6.3 Revenue Model 14
6.4 Cost Estimation 15
6.5 Pricing Strategy 16
6.6 Competitive Advantage 16
6.7 Intellectual Property Considerations 16
7. BUDGET AND JUSTIFICATION 17
7.1 Overall Justification 17
8. WORK BREAKDOWN STRUCTURE (WBS) 18
8.1 Timeline Alignment 18
8.2 Balanced Workload Distribution 18
9. GANTT CHART 19
10. REFERENCES 20
11. APPENDICES 21

LIST OF ABBREVIATIONS

API	Application Programming Interface
Brotli	Brotli Compression Algorithm
FR	Functional Requirement
GZIP	GNU Zip Compression Format
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
Mbps	Megabits per second
ms	Milliseconds
NFR	Non-Functional Requirement
O/L	Ordinary Level
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UI	User Interface
WBS	Work Breakdown Structure

1. **INTRODUCTION**

1.1 Background and Context

Web-based learning systems increasingly rely on real-time data exchange to provide interactive and personalized educational experiences. In systems that incorporate emotion-aware features, communication between the client and server becomes even more important because learner-related data, including emotional state information, interaction records, and adaptive support requests, must be transmitted efficiently and securely. Within the broader project, the proposed platform is intended to support Deaf G.C.E. O/L Information Technology students in Sri Lanka through an accessible emotion-aware e-learning environment. As defined in the approved project context, the overall system integrates emotion recognition, adaptive learning support, and chatbot-based interaction for Deaf learners.

Although the main platform addresses accessibility and intelligent learning support, the effectiveness of such a system also depends on how efficiently and safely data is communicated across the network. Real-time educational applications can experience performance degradation when network bandwidth is limited, latency is high, or communication protocols are not optimized. This issue is especially relevant in low-connectivity educational environments, where unstable or constrained internet access can reduce responsiveness and negatively affect user experience. In addition, because the system handles sensitive learner-related information, including emotion-associated data, secure transmission is essential to protect privacy and support ethical use of the platform. The proposal template also expects the Introduction to explain the real-world problem, affected users, importance of the issue, and the proposed direction of the solution.

From the perspective of this individual contribution, the key problem is not the development of the full learning platform, but the lack of a secure and bandwidth-efficient communication approach that can support real-time interaction in the proposed web application. Conventional implementations often prioritize functionality at the application layer while giving limited attention to transmission efficiency, network overhead, and in-transit protection of sensitive data. As a result, even well-designed intelligent learning features may become less effective when deployed in real-world educational settings with constrained connectivity or insufficient transport-layer safeguards. Therefore, this individual research focuses on designing and evaluating a communication framework that improves both network performance and data protection for the proposed system. This direction is also consistent with the proposal's guidance that individual work should remain feasible, measurable, and directly linked to a clearly defined technical problem.

1.2 Literature Review

Existing research in web-based educational systems, real-time interactive applications, and intelligent learning environments has shown increasing emphasis on personalization, accessibility, and responsiveness. Many studies focus on learner-centered features such as adaptive content delivery, chatbot-based assistance, and real-time emotion recognition to improve engagement and learning outcomes. At the same time, secure communication practices such as HTTPS/TLS, access control, and protected API communication are widely recognized as essential for safeguarding user data in web applications. Similarly, network optimization techniques such as payload reduction, response compression, and efficient client-server communication are commonly used to improve performance in data-driven systems. However, these areas are often treated independently, with research typically emphasizing either application intelligence or general web security rather than evaluating how secure and optimized transmission affects real-time educational interaction. This is important because, according to the project structure, the success of the overall system depends not only on intelligent features but also on the efficient and secure movement of learner-related data between components.

Furthermore, most accessible e-learning research and implementations primarily address content accessibility, user interaction, and interface design, while giving comparatively less attention to the communication layer that supports these services. In real-time systems involving emotion-related processing, transmission inefficiency can increase latency and bandwidth consumption, while weak transport security may expose sensitive learner information during communication. The lecture and proposal guidance emphasize that a strong proposal must identify a concrete technical problem, justify it with evidence, and define a measurable evaluation strategy rather than presenting only a general development idea. Therefore, for this individual contribution, the relevant literature domain is the intersection of secure web communication, bandwidth optimization, and real-time educational data exchange. This perspective positions the research within a specialized networking and security context rather than the broader application development domain.

1.3 Research Gap

Although prior work has addressed accessibility, emotion-aware learning, adaptive educational systems, and secure web communication as separate areas, there is limited evidence of an experimentally evaluated communication framework that specifically balances data transmission efficiency and in-transit protection for a real-time emotion-aware educational web application. Existing systems commonly prioritize application-level functionality without sufficiently analyzing how payload size, compression strategies, encrypted communication, and network constraints affect responsiveness and scalability in low-connectivity learning environments. This reveals a clear gap in the design and evaluation of privacy-aware, bandwidth-optimized communication models for such systems. Accordingly, this individual research addresses that gap by proposing and evaluating a secure and optimized client-server communication framework for the broader emotion-aware e-learning platform, with emphasis on latency, throughput, bandwidth consumption, and security overhead under constrained network conditions. This gap definition also aligns with the proposal expectations that the individual contribution should be measurable, technically justified, and directly connected to the research methodology and system design.

1.4 Research Problem

Network security and optimization are widely used in today's communication systems. For a system which uses real-time facial data to calibrate in real-time to meet user requirements is very sensitive with communication. Therefore, having better communication is one of major requirements in those systems. Even though the systems come up with better security and communication, in a country like Sri Lanka where network connectivity has bandwidth issues. It is very useful to have better network architecture to a system to perform well under constraint network conditions. It will be very useful for students who are in rural areas with less bandwidth network connections.

Therefore, in my project I am doing research to overcome these problems and come up with a suitable method to perform secure and optimized network architecture for the system.

2. OBJECTIVE

2.1 Main Objective

To develop and experimentally validate a secure and bandwidth-efficient communication framework for the proposed emotion-aware e-learning web application, with the aim of ensuring protected and low-latency transmission of sensitive learner-related data in constrained network environments.

2.2 Specific Objectives

- To design a secure client–server communication model for transmitting learner-related data within the proposed web application.
- To optimize network data transmission through payload minimization and HTTP compression techniques to improve bandwidth efficiency.
- To ensure secure in-transit data protection using HTTPS/TLS and network-level access control mechanisms.
- To conduct controlled experiments to measure latency, throughput, and bandwidth consumption under different network conditions.
- To evaluate the trade-off between communication security and transmission performance by comparing baseline and optimized communication approaches.

3. METHODOLOGY

This adopts an experimental research methodology combined with system design and evaluation to investigate how secure and bandwidth-optimized communication can improve data transmission in the proposed emotion-aware e-learning web application. The research focuses on the communication layer of the system, specifically on how learner-related data can be transmitted efficiently and securely between the client browser and the backend server under both normal and constrained network conditions. This methodological choice is appropriate because the objective of this study is not to develop the full application, but to design a communication framework and evaluate its effectiveness using measurable network and security performance indicators. Such an approach is consistent with the proposal guidance that methodology must directly support the stated objectives and generate evidence for evaluation.

The study will follow a comparative experimental design. Two communication scenarios will be evaluated: a baseline communication approach, representing standard client–server transmission without optimization, and a proposed optimized secure communication approach, which incorporates payload reduction, HTTP compression, and secure transport mechanisms. By comparing these two scenarios, the study will identify whether the proposed framework improves communication efficiency while maintaining protection of sensitive learner-related data. This comparison is necessary to evaluate the trade-off between communication performance and security overhead, which forms the main research contribution of this component.

From a technical perspective, the proposed framework will apply several methods. HTTPS/TLS will be used to secure communication between the browser and the server to protect learner-related data during transmission. Payload optimization will be applied by minimizing unnecessary data fields and structuring requests in a more efficient way. In addition, HTTP compression techniques, such as GZIP or Brotli, will be used to reduce the size of transmitted data and improve bandwidth efficiency. Basic network-level protection mechanisms, including access control and restricted communication endpoints, will also be considered to strengthen transport-layer security. These methods are selected because they are practical, measurable, and directly related to the problem of ensuring secure and efficient communication in real-time web systems.

The implementation and evaluation environment will be based on a web application context in which client requests and server responses can be monitored and measured. The study will use appropriate tools for traffic inspection, request analysis, and communication testing, such as browser developer tools, API testing tools, and network monitoring utilities. The selected tools will help capture request size, response time, bandwidth usage, and communication behavior under different transmission settings. These platforms are suitable because they provide direct evidence for evaluating the effectiveness of the proposed communication framework without requiring changes to the core application logic developed by other team members.

Data collection in this study will be based primarily on quantitative experimental measurements. The main performance indicators will include latency, throughput, bandwidth consumption, and security overhead introduced by encrypted communication and compression techniques. Experiments will be conducted under both normal and simulated low-bandwidth conditions to reflect realistic deployment constraints relevant to educational environments. The collected results will be recorded and analyzed using comparative tables, graphs, and percentage-based performance differences between the baseline and proposed approaches. Since the communication layer involves learner-related and emotion-associated data, ethical considerations will also be considered by emphasizing secure transmission, minimal exposure of sensitive data, and avoidance of unnecessary data retention during testing.

The validation strategy of this study is based on determining whether the proposed framework provides measurable improvement over the baseline communication approach. A successful outcome will be indicated by reduced bandwidth consumption, lower or acceptable latency, improved transmission efficiency, and secure encrypted communication with manageable overhead. By combining system design with controlled experimental evaluation, this methodology enables the study to produce objective evidence on the effectiveness of a secure and optimized communication framework for the broader emotion-aware e-learning platform. This aligns with the expectation that an individual proposal should present a feasible, measurable, and technically justified research plan rather than only a development intention.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution

The proposed Emotion-Aware E-Learning System is designed to support deaf students studying the G.C.E. O/L Information Technology syllabus through an intelligent and adaptive learning environment. The system integrates multiple Artificial Intelligence technologies including Natural Language Processing, facial emotion recognition, and adaptive learning algorithms to provide personalized learning support.

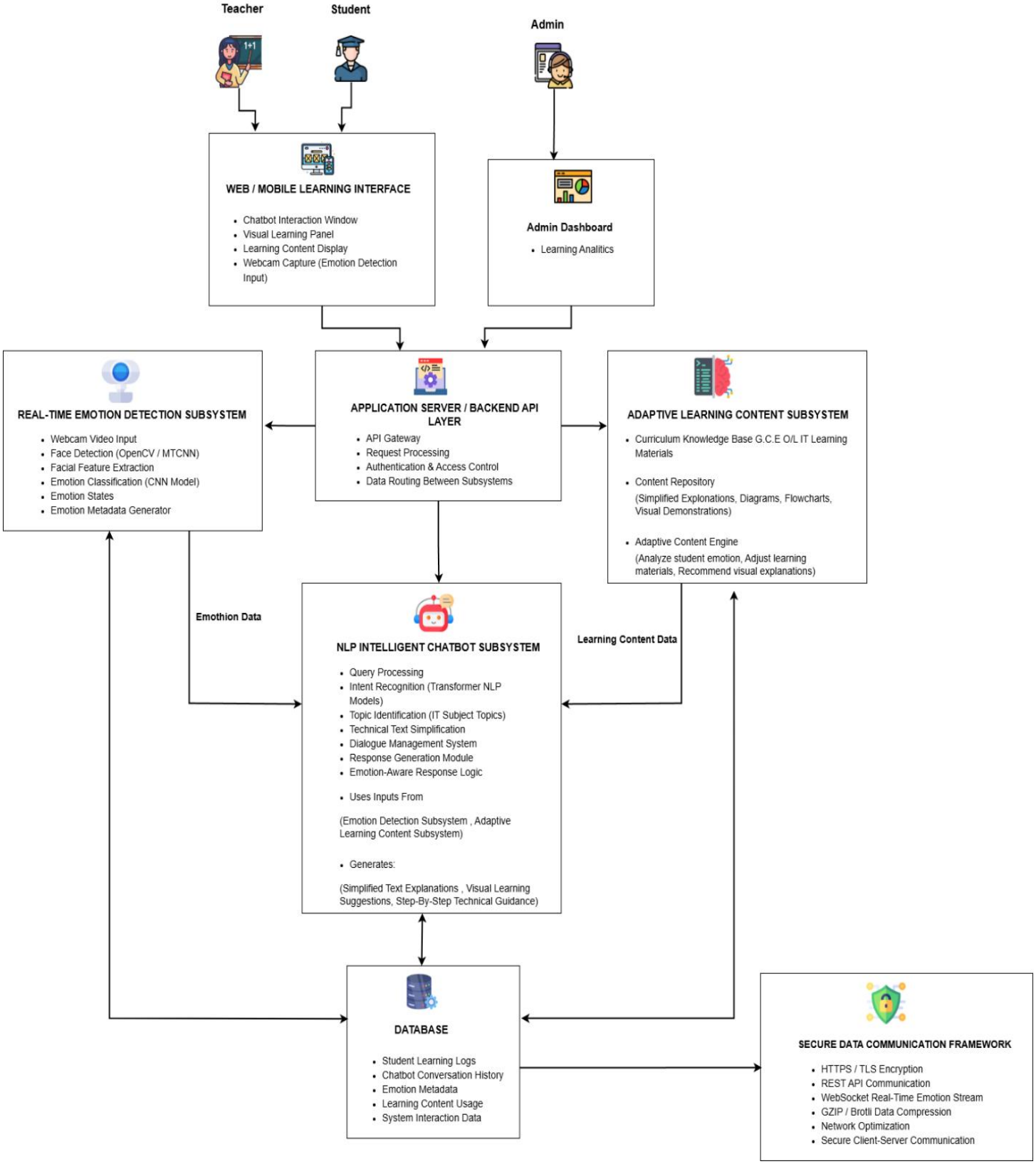
The architecture consists of three major intelligent subsystems: the Real-Time Emotion Detection Subsystem, the Adaptive Learning Content Subsystem, and the NLP Intelligent Chatbot Subsystem. These subsystems operate together through a centralized Application Server / Backend API Layer, which manages communication, data processing, and integration between the different components of the system.

The Emotion Detection Subsystem analyzes the student's facial expressions captured through the webcam to detect emotional states such as confusion, frustration, or engagement. Based on these emotional signals, the Adaptive Learning Content Subsystem dynamically modifies the learning materials by providing simplified explanations, visual diagrams, and step-by-step demonstrations. The NLP Intelligent Chatbot Subsystem acts as an interactive tutor that answers student questions by combining the student's query, emotional state, and learning context to generate appropriate responses.

Additionally, the system includes a secure data communication framework that ensures safe and efficient transmission of information between the client interface and the backend server using encrypted communication protocols.

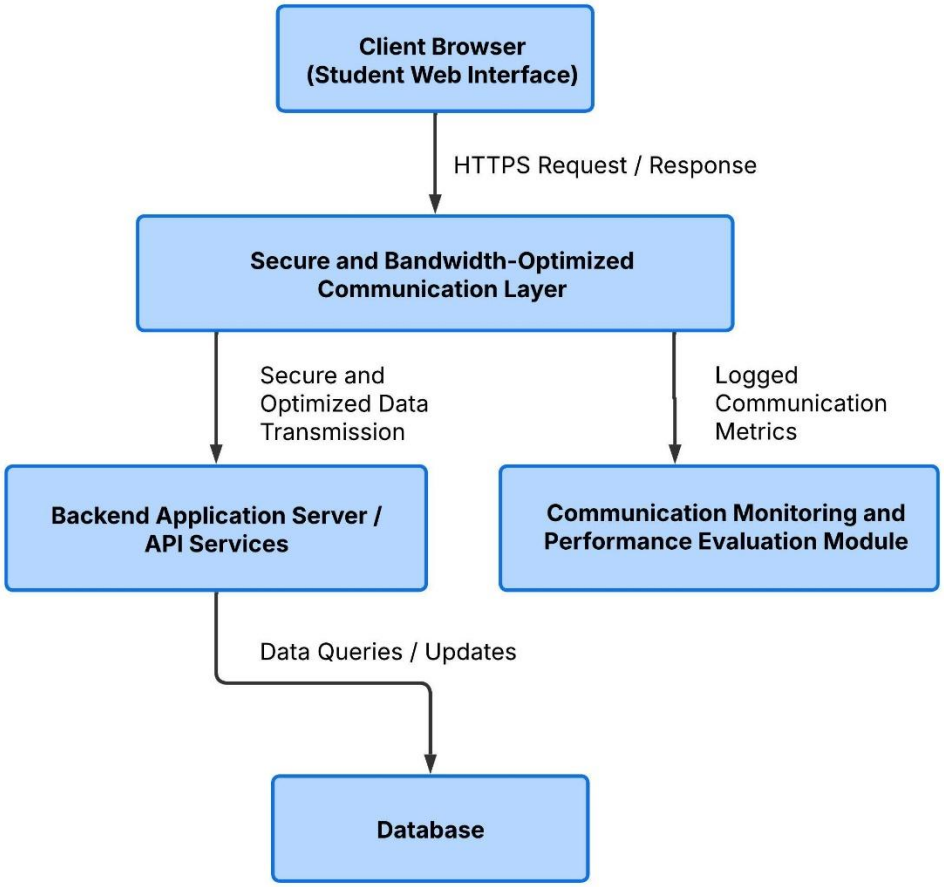
4.2 High-Level System Architecture Diagram

Figure 1 High-Level System Architecture Diagram



4.3 **Individual Components and** Functions

Figure 2 Individual Components



1. Secure Transport Module

which is responsible for protecting data during transmission using HTTPS and TLS-based encryption. This module ensures that learner-related data exchanged between the browser and the backend cannot be easily intercepted or exposed while traveling across the network.

2. Payload Optimization Module

which focuses on reducing communication overhead by minimizing unnecessary request data, structuring payloads efficiently, and applying HTTP compression techniques such as GZIP or Brotli. This is especially important in low-bandwidth environments, where large payloads can increase delays and negatively affect responsiveness.

3. Access Control and Protection Module

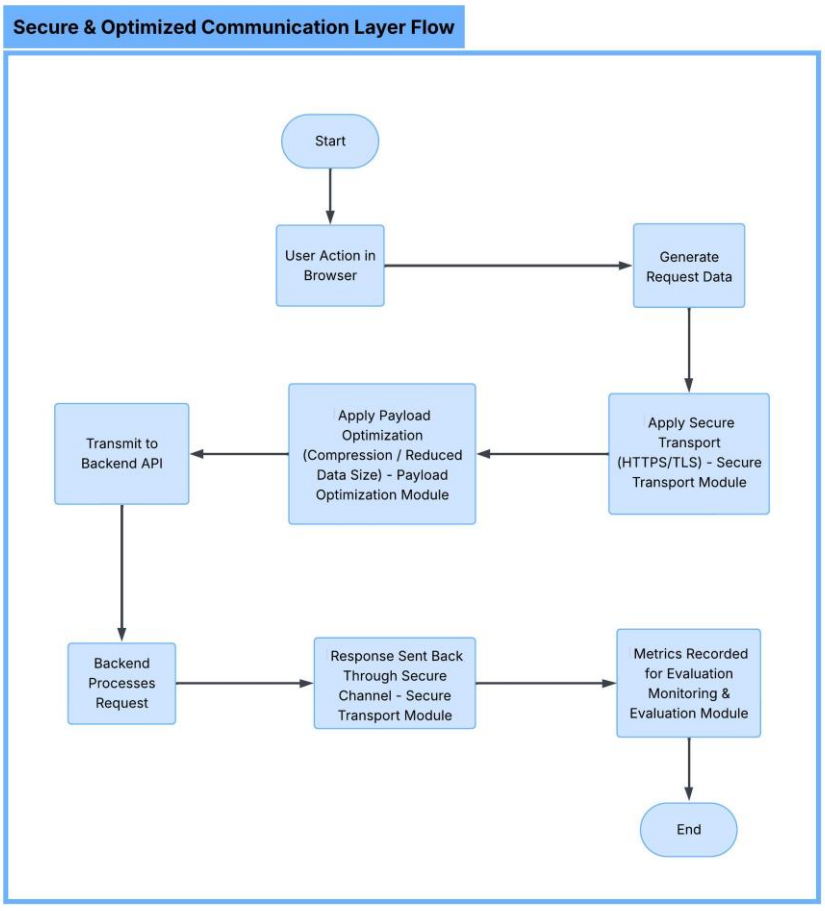
which strengthens the communication layer through protected endpoints, restricted access policies, and basic network-level security controls.

4. Monitoring and Evaluation Module

which records transmission-related measurements such as request size, latency, and bandwidth usage for later experimental analysis.

4.4 Data Flow Explanation

Figure 3 Individual Component Data Flow



From a data flow perspective, the communication layer handles several forms of learner-related traffic that may be exchanged in the broader system, including learner interaction data, emotion-related metadata, chatbot or support requests, and application responses. The data flow begins at the client's browser, where a request is generated.

The request is then protected using HTTPS/TLS, optimized to reduce unnecessary transmission overhead transmission, and forwarded to the backend API. After processing, the backend server returns a response to the client, again through the same protected and optimized channel. This design allows the system to support real-time interaction while reducing communication inefficiencies. It also creates a measurable pathway through which network-related performance indicators can be captured and compared under different conditions.

4.5 Scalability and Security Considerations

Scalability and security are key considerations in this architecture. From a security perspective, the design emphasizes encrypted in-transit communication, controlled access to endpoints, and reduced exposure to sensitive learner-related data. This is important because the broader system processes user interaction and emotion-associated information that should not be transmitted insecurely.

From a scalability perspective, reducing payload size and applying compression can lower bandwidth consumption and improve responsiveness, especially when multiple users interact with the system or when internet quality is limited. As a result, the proposed communication architecture not only improves transport-layer protection but also enhances the suitability of the overall platform for real-world educational deployment in constrained network environments such as Sri Lanka.

Overall, the high-level system architecture of this individual contribution demonstrates how a secure and bandwidth-optimized communication layer can be integrated into the proposed emotion-aware e-learning web application. By focusing on secure transport, payload optimization, access protection, and measurable communication performance, the architecture directly supports the objectives of this study and provides a clear basis for subsequent experimental evaluation. This also follows the proposal guidance that the architecture should reflect the research objective, show major interactions and data flow, and remain feasible within the scope of the individual contribution.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders relevant to this individual component are the **student users**, the **system administrators/developers**, and the **research team**. Student users interact with the web application through the browser, and although they do not directly operate the communication layer, they are affected by its performance, reliability, and security. System administrators and developers are stakeholders because they configure, maintain, and integrate the communication framework with the backend services. The research team is also a stakeholder, as the communication layer must provide measurable performance data for experimental evaluation and validation.

In addition, educational institutions that may deploy the system are indirect stakeholders, because they benefit from a communication framework that is secure, bandwidth-efficient, and suitable for low-connectivity learning environments. Therefore, this individual component must be designed not only for technical functionality, but also for reliability, usability, and deployment feasibility within a real educational context.

5.2 Functional Requirements

The proposed communication component shall satisfy the following functional requirements:

FR1. The system shall establish secure client–server communication between the student web interface and the backend server.

FR2. The system shall transmit learner-related requests and responses through an encrypted HTTPS/TLS channel.

FR3. The system should support payload optimization by reducing unnecessary transmitted data fields.

FR4. The system shall support HTTP compression mechanisms such as GZIP or Brotli for eligible transmitted data.

FR5. The system shall allow secure access to backend API endpoints through controlled communication paths.

FR6. The system shall log communication-related performance data, including request size, response time, and bandwidth usage, for evaluation purposes.

FR7. The system shall support transmission of learner-related metadata and interaction requests required by the broader web application.

FR8. The system shall provide a communication pathway that can be experimentally tested under different network conditions.

5.3 Non-Functional Requirements

The proposed communication framework satisfies the following non-functional requirements:

5.3.1 Security Requirements

NFR1. All data transmitted between the client's browser and backend server shall be protected using HTTPS/TLS for encryption.
NFR2. Sensitive learner-related and emotion-associated data shall not be exposed through unsecured transport channels.
NFR3. Access to backend communication endpoints shall be restricted to authorized requests only.

5.3.2 Performance Requirements

NFR4. The communication framework shall reduce transmission overhead through payload optimization and compression.
NFR5. The system shall maintain acceptable response time under normal and constrained bandwidth conditions.
NFR6. The communication layer shall support efficient bandwidth usage for real-time interaction within the web application.

5.3.3 Reliability Requirements

NFR7. The communication layer shall provide stable request-response behavior during continuous web-based interaction.
NFR8. Communication monitoring and logging shall operate consistently during experimental evaluation.

5.3.4 Scalability Requirements

NFR9. The communication framework shall support increasing client requests without significant degradation of transmission efficiency.
NFR10. The architecture shall remain suitable for deployment in low-connectivity educational environments.

5.3.5 Maintainability Requirements

NFR11. The communication layer should be modular enough to integrate with the broader application without requiring major changes to other members' components.
NFR12. The framework shall allow future modification of optimization and security configurations without redesigning the full system.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed Emotion-Aware E-Learning System includes educational institutions and organizations that support deaf and hard-of-hearing students. This includes special education schools, inclusive schools, and educational training centers that provide learning support for students with hearing impairments.

The system is specifically designed for students studying the G.C.E. Ordinary Level (O/L) Information Technology syllabus**,** where technical concepts often require clear explanations and visual learning materials.

A typical customer persona is a 15–16-year-old deaf student preparing for the G.C.E. O/L Information Technology examination**.** These students prefer visual learning methods such as diagrams, simplified explanations, and step-by-step demonstrations instead of long text-based lessons.

Secondary users include teachers and school administrators who want to improve learning accessibility and monitor student learning progress through analytics.

6.2 Value Proposition

The proposed system provides an intelligent learning platform specifically designed to improve accessibility for deaf students. Traditional e-learning systems rely heavily on text or audio explanations, which can be difficult for deaf learners to understand.

This platform offers several key benefits:

- **Emotion-aware learning support** that detects student emotions during learning sessions.
- **Adaptive learning content** that adjusts explanations and visual materials based on the student's understanding.
- **Interactive AI chatbot tutor** that answers student questions using simplified explanations.
- **Visual learning materials** such as diagrams, flowcharts, and step-by-step demonstrations.

By combining emotion detection, adaptive learning, and NLP-based chatbot support, the system creates a more personalized and inclusive learning environment for deaf students.

6.3 Revenue Model

The system can generate revenue through several simple models suitable for an educational platform.

- **Subscription Model:**
  Educational institutions can subscribe to the platform by paying a yearly or monthly fee to allow their students to use the system.
- **Institutional Licensing:**
  Schools or educational centers can purchase a license that allows multiple students and teachers to access the system.
- **Freemium Model:**
  Basic learning support and chatbot features can be provided for free, while advanced features such as detailed learning analytics, additional visual resources, and advanced AI support can be included in a premium plan.

6.4 Cost Estimation

The commercial deployment of the system may involve costs related to cloud hosting, domain and SSL services, maintenance, monitoring, technical support, and future scaling of server resources. Additional cost factors may include training, customization, and institutional onboarding. For my individual component, the main relevant cost areas include secure hosting infrastructure, encrypted communication support, and monitoring tools required for performance analysis and maintenance.

Although security and performance optimization may slightly increase initial technical setup cost, these investments improve long-term reliability and reduce the risk of data exposure, service instability, and inefficient bandwidth usage. Therefore, the communication framework contributes positively to the cost-effectiveness of large-scale deployment.

Table 1 Cost Estimation

Category	**Estimated Cost (Initial Year)**	**Description**
Development Tools	Low / Free	Use of open-source frameworks such as Python, OpenCV, TensorFlow, and NLP libraries
Cloud Hosting	$60 – $80	Basic cloud server for backend API and database
Learning Content Production	$20 - $50	Creating diagrams and visual learning materials
Database Services	$30 – $50	Hosting MongoDB database
System Maintenance	$50 - $80	System updates and bug fixes

6.5 Pricing Strategy

The pricing strategy focuses on affordability so that educational institutions and students can easily adopt the system.

Possible pricing options include:

1. **Free Basic Plan:** Limited chatbot interaction and basic learning support.

2. **Student Plan:** Approximately $3 - $5 per month for full access to chatbot tutoring and adaptive learning materials.

3. **Institutional Plan:** Approximately $200 - $500 per year for schools that want to provide access to multiple students.

This pricing model ensures that the platform remains accessible while still providing sustainable funding for system maintenance and improvements.

6.6 Competitive Advantage

The proposed system offers several advantages compared with traditional e-learning platforms.

First, the system is specifically designed for deaf and hard-of-hearing students**,** focusing on visual explanations and simplified learning content.

Second, the integration of emotion detection technology allows the system to understand student emotional states and adjust learning materials accordingly.

Third, the system combines emotion detection, adaptive learning, and chatbot tutoring on a single platform, which is not commonly available in traditional e-learning systems.

These features provide a more personalized and engaging learning experience for students.

6.7 Intellectual Property Considerations

The proposed system includes several components that can be protected as intellectual property.

The software architecture, chatbot algorithms, and system implementation can be protected through software copyright protection**.** Additionally, custom-developed learning materials, diagrams, and educational content can also be protected under copyright laws.

If the system demonstrates significant novelty, future research may explore applying for **p**atent protection for the emotion-aware adaptive learning methodology used in the platform.

Protecting the intellectual property of the system will help ensure ownership of the developed technology and support potential commercialization opportunities in the future.

7. BUDGET AND JUSTIFICATION

Table 2 Budget

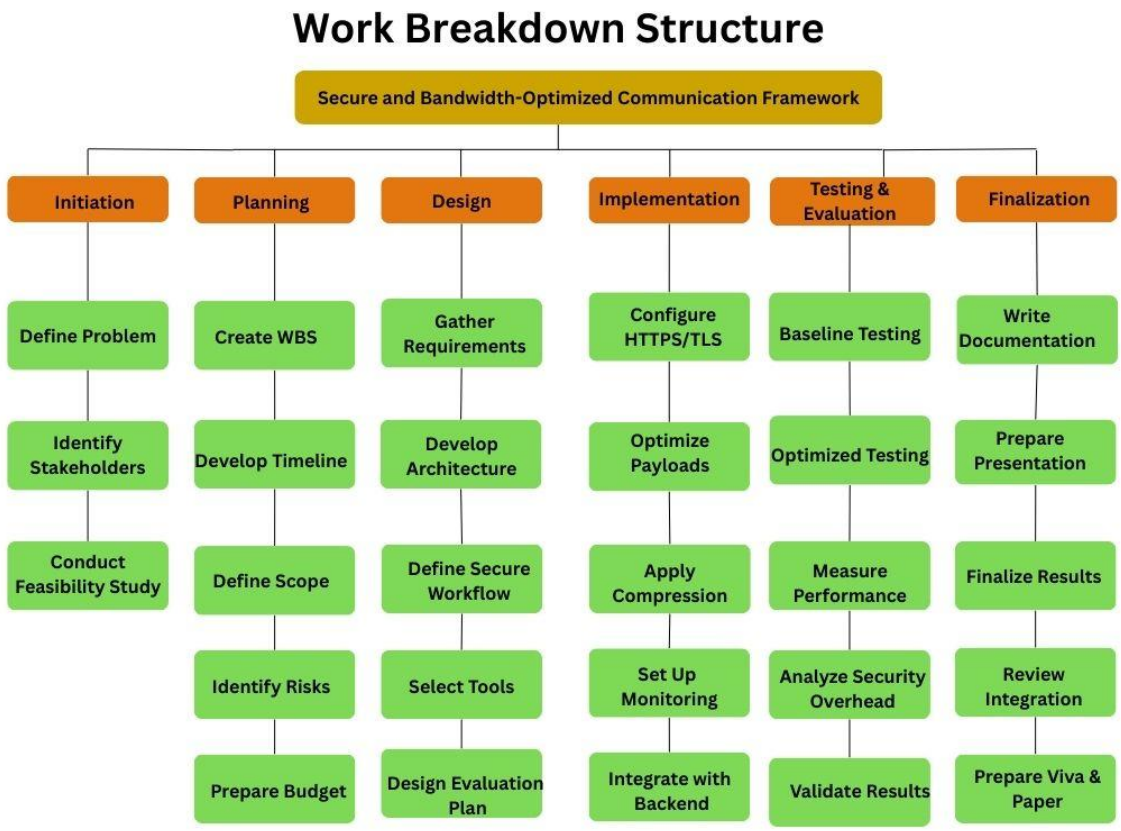
Category	Item	Estimated Cost in LKR
Hardware	Existing Personal Laptop	0.00
Hardware	Internet Connectivity	6000.00
Software	Open-source Tools	0.00
Cloud Services	Hosting Environment	15000.00
Human Charges	Secure Communication Framework Design, Implementation, Testing, and Evaluation	0.00
Total Estimated Budget		21000.00

7.1 Overall Justification

The proposed budget is reasonable and feasible for an undergraduate research project focused on secure web communication and network optimization. By using an existing personal computer and open-source tools, the study minimizes unnecessary cost while still supporting realistic implementation and evaluation. The main expenses are associated with internet usage, cloud-based testing support, and the human effort required to configure, optimize, and experimentally validate the proposed communication framework. This low-budget approach also demonstrates that secure and bandwidth-efficient communication can be studied and implemented in a practical and cost-sensitive manner.

8. WORK BREAKDOWN STRUCTURE (WBS)

Figure 4 Work Breakdown Structure



8.1 Timeline Alignment

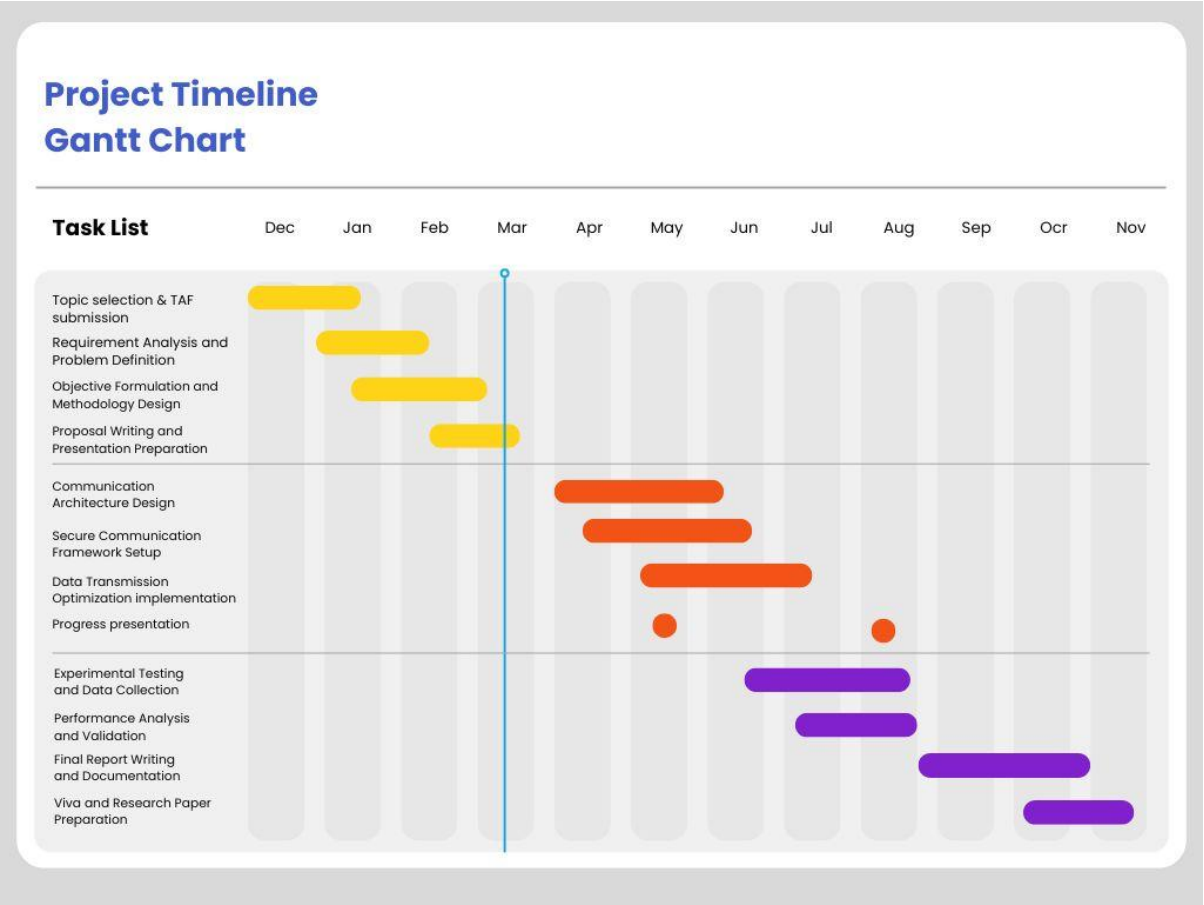
Early stages of the project are mainly focused on requirement analysis, literature review, architecture design, and secure framework planning. The middle stage is allocated for implementation of communication optimization and monitoring mechanisms. This sequencing ensures that the work remains feasible and supports timely progress toward proposal submission, progress evaluations, final report submission, and viva preparation.

8.2 Balanced Workload Distribution

The workload of this individual component is balanced across design, implementation, evaluation, and documentation activities. Initial effort is allocated to academic planning and system design, ensuring that the contribution is properly scoped and justified. The central part of the workload is devoted to the technical configuration of secure communication and transmission optimization. Finally, report writing and presentation preparation are included to ensure that the findings of the component are properly documented and communicated. This balanced distribution helps maintain feasibility within the available timeline and ensures that the component makes a measurable and independent contribution to the overall project.

9. GANTT CHART

Figure 5 Gantt Chart



10. REFERENCES

[1] M. M. Bala, H. Akkineni and C. Srinivasulu, "An Approach for Learner Categorization Based on Emotions in Intelligent Adaptive E-Learning Environment," in Journal of Mobile Multimedia, vol. 18, no. 6, pp. 1709-1732, November 2022.

[2] A. Mohamed et al., "Emotion-Aware Course Personalization in Virtual Classrooms via Multi-Modal Fusion Learning Systems," 2025 3rd International Conference on Cyber Resilience (ICCR), Dubai, United Arab Emirates, 2025, pp. 1-7.

[3] M. L, E. Anbalagan, M. Y. Al-Safarini, Y. A. Satyavati and V. H. Raj, "Emotion-Aware Sentiment Analysis in User-Generated Content via Attention-Augmented Deep Learning Frameworks," 2025 IEEE 6th Global Conference for Advancement in Technology (GCAT), Bangalore, India, 2025, pp. 1-6

[4] W. H. K. Fernando, P. L. D. N. Perera, M. I. N. Fernando, P. T. E. W. Gunasekara, H. Fernando and R. De Zoysa, "Empowering Deaf Children with Sinhala Sign Language, Emotion Detection, and Sound Recognition," 2023 5th International Conference on Advancements in Computing (ICAC), Colombo, Sri Lanka, 2023, pp. 388-393.

[5] D. J.M.D.G.K.M, D. G.A.I.P, H. S.N.N, D. W.W.M.R.T.I.W, D. R. Wijendra and S. Kasthuriarachchi, "Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka," 2023 5th International Conference on Advancements in Computing (ICAC), Colombo, Sri Lanka, 2023, pp. 846-851.

[6] C. Sathiyapriyan, B. Manikandan, S. Gokul, R. Satheesh, S. Nivetha and S. Leelavathy, "Strategies for Building and Maintaining Secure Web Applications," 2024 5th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI), Lalitpur, Nepal, 2024, pp. 803-809.

[7] M. Nagpal, G. Rajalakshmi, N. Vashisht, N. Beemkumar, J. V. Bagade and K. Yuvaraj, "Data Compression Techniques for High-Speed NetworkCommunications," 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India, 2024, pp. 1-6.

[8] L. M. I. L. Joseph, R. Rallan, S. Agarwal, S. S. Shrivastava, G. D and A. P. Nayaki, "Edge-EmoNet: A Low-Latency Edge-based Deep Learning Framework for Real-Time Emotion Recognition with Attention-Optimized Feature Fusion," 2025 7th International Conference on Innovative Data Communication Technologies and Application (ICIDCA), Coimbatore, India, 2025, pp. 631-637.

[9] "IEEE Standard for Data Processing and Compression Framework for Internet of Things," in IEEE Std 3122-2025, vol., no., pp.1-21, 28 Nov. 2025.

[10] Z. Sun, B. Peng and L. Chen, "Research on Secure Transmission System Based on Data Encryption of New Energy Station," 2024 4th Power System and Green Energy Conference (PSGEC), Shanghai, China, 2024.

[11] P. Kumar, R. Ranjan, D. Goyal, G. Srinivas and B. Singh, "Real-Time Steganographic Tool for Secure Data Transmission in Live Video Streams Using AES Encryption," 2025 International Conference on Engineering, Technology & Management (ICETM), Oakdale, NY, USA, 2025.

11. APPENDICES

Appendix A - High-Level Architecture Diagram

This appendix contains the high-level architecture diagram of the proposed secure and bandwidth-optimized communication framework and its placement within the broader web-based learning system.

Appendix B - Individual Component Diagram

This appendix contains the internal component diagram of the proposed communication framework, including the Secure Transport Module, Payload Optimization Module, Access Control and Protection Module, and Monitoring and Evaluation Module.

Appendix C - Data Flow Diagram

This appendix presents the data flow diagram of the individual communication component, showing how learner-related requests are protected, optimized, transmitted, processed, and measured during communication between the client and backend server.

Appendix D - Work Breakdown Structure (WBS) Chart

This appendix includes the phase-based Work Breakdown Structure chart of the individual component, covering initiation, planning, design, implementation, testing and evaluation, and finalization activities.

Appendix E - Gantt Chart

This appendix contains the Gantt chart prepared for the individual communication and security component, aligned with the official project timeline and major academic milestones.

Appendix F - Preliminary Testing and Evaluation Plan

This appendix includes the initial evaluation plan for the proposed communication framework, including the baseline and optimized communication scenarios, performance metrics, and testing conditions.

Appendix G - Risk Considerations

This appendix summarizes possible technical and project-related risks associated with the communication component, including implementation delays, configuration challenges, testing limitations, and mitigation strategies.